

Department of Computer Science and Engineering
(Punjabi University, Patna)

Subject: Basic Electrical Engineering

Max. Marks: 15

Time: 60 Minutes

Section-A (1x5) (All questions are compulsory)

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Q1

- ✓ a) State the losses in a transformer? (1)
- b) Write the electrical equivalent of mechanical power developed in the armature of a DC motor? (1)
- c) State the difference between DC generator and DC motor? (1)
- d) What is the effect of armature reaction on the main pole flux in a DC generator? (1)
- ✓ e) Draw the phasor diagram of an ideal transformer on inductive load? (1)

Section-B (2x5) (Attempt any two questions)

Q2 (a) Derive the EMF equation of a transformer? (3)

(b) What is the role of commutator in a DC generator? (2)

Q3 (a) When a 100 kVA, single phase transformer was tested, the following results were obtained:

On open-circuit the power consumed was 1300 W and on short circuit at full-load current the power consumed was 1200 W.

Calculate the efficiency of transformer on full load at unity power factor? (2)

(b) How open-circuit (or no-load test) test is performed on a transformer? Draw its set up and discuss the results obtained from the test? (3)

✓ Draw and explain characteristic curves for DC series motor showing relationships between (i) speed and armature current/load, (ii) torque and armature current, (iii) speed and torque? (5)

(MST II) Department of Computer Science and Engineering
(Punjabi University, Patiala)

Subject: Basic Electrical Engineering (ECEB1203T)

Max. Marks: 15

Time: 60 Minutes

Section-A (5x1) (All questions are compulsory)

Q1

- a) State the significance of back emf in DC motor? (1)
- b) What is operating principle of transformer? (1)
- c) Write the condition for maximum efficiency of a transformer? (1)
- d) Write the EMF equation of a transformer? (1)
- e) Draw the phasor diagram of an ideal transformer on pure resistive load? (1)

Section-B (2x5) (Attempt any two questions)

- Q2) Draw and explain characteristic curves for DC shunt motor showing relationships between (i) speed and armature current/load, (ii) torque and armature current, (iii) speed and torque? (5)
- Q3) (a) Derive the Torque equation of a DC motor? (3)
(b) What is Transformation Ratio? How transformers are classified on the basis of it? (2)
- Q4) How Short-Circuit test is performed on a transformer? Draw its set up and discuss the results obtained from the test? (5)

(MST II) Department of Computer Science and Engineering
Subject: Basic Electrical Engineering (ECEB1103T)

Max. Marks: 15

(Punjabi University, Patiala)
Time: 60 Minute

Section-A (5x1) (All questions are compulsory)

Q1

- a) Tell the properties/assumptions of an ideal transformer? (1)
- b) What is the operating principle of a transformer? (1)
- c) Write the condition for maximum efficiency of a transformer? (1)
- d) Write the Torque equation of a DC motor? (1)
- e) State the difference between DC generator and DC motor? (1)

Section-B (2x5) (Attempt any two questions)

Q2) Discuss various types of DC Generators? (5)

Q3) (a) Derive the EMF equation of a transformer? (3)

(b) Discuss the phasor diagram of an ideal transformer on inductive load? (2)

Q4) (a) Why Short-Circuit test is performed on a transformer? (2)

(b) Draw and explain speed versus armature current characteristic curve for DC series motor?